EV Charging Demand Forecasting Documentation

Executive Summary

This project forecasts EV charging demand, both session and energy consumption, to support staffing, capacity planning, and energy procurement decisions. Three forecasting approaches (Prophet, SARIMA, XGBoost) were built and compared under fair, forecast-realistic conditions. XGBoost, using only calendar information, was the most accurate model and was used to generate 30/90/365-day forecasts, visualised in an interactive Power BI dashboard. Two findings shaped the project: a systematic data quality issue in the source data and a genuine disagreement between models about whether 2025 demand will keep growing or plateau. Both are explained transparently.

1. The Data and Key Quality Findings

The dataset covers 290,024 charging sessions across 34 stations and 199 chargers, from January 2022 to December 2024. The data was largely clean, with no missing values or duplicates, and all numeric fields within plausible ranges.

One significant issue was found and highlighted as a deliberate part of the analysis. Cross-checking session timestamps against each charger's recorded installation date revealed that 30% of all sessions (88,466 rows) occurred before the charger was supposedly installed, and this wasn't isolated to a few records. Every single charger in the dataset showed the issue to some degree, and the severity scaled almost linearly with how late in 2022 the charger came online (correlation ≈ 0.78). That pattern is a data generation artifact. Session data appears to have been generated across a fixed calendar range independent of each charger's actual install date, rather than scattered bad records.

Because this project's forecasting task does not depend on installation date, the field was simply excluded from modelling rather than dropping the affected rows, which would have unevenly stripped data from later-installed chargers and biased the resulting demand series. The finding is flagged here because it would matter for any future projects that do rely on that field.

2. What Demand Actually Looks Like

Two demand measures were built: session count (for staffing/capacity) and total energy in kWh (for procurement), aggregated at both hourly and daily resolution.

Three patterns stood out:

- **A strong, repeated daily rhythm**

Demand is near-zero overnight, climbs from 5 am, plateaus from late morning through early afternoon, and tapers through the evening. It follows a daytime/workplace charging pattern, not an overnight home charging one.

- **A clear weekday effect**

Weekday demand runs roughly 27% above weekends, reinforcing that this is commuter-driven usage.

- **Most of the historical growth in total demand is network expansion, not rising demand per station**

Total session volume jumped sharply right at January 2023 and January 2024, almost exactly matching when 9 and then 16 new stations came online. This distinction matters directly for capacity planning: it means the network isn't necessarily getting busier per station, just bigger.

3. Building and Comparing Three Forecasting Models

Each model was trained on the first ~ 2.7 years of data and evaluated against a held-out final 90 days it never saw during training.

Prophet initially performed badly (RMSE ≈ 60.6) because its default settings assumed seasonal swings stay a fixed size, an assumption that breaks when the network itself is growing. Switching to multiplicative seasonality, where swings scale with the trend, cut error roughly by 8x.

SARIMA tracked the timing of the daily cycle well but consistently underestimated how high demand actually peaked. This was a distinct limitation from Prophet's, and was ultimately left behind in the final comparison.

XGBoost was tested in two days. An initial version using actual values (what happened 1 hour, 1 day, and 1 week prior) produced the lowest error of any model, but relied on information that does not exist when forecasting genuinely future dates, so it was kept only as a reference point. A second, calendar-only version (hour, day to week, month, and year without requiring real-time data) was built specifically for actual forecasting and still outperformed both Prophet and SARIMA.

Final comparison, all models evaluated under identical, forecast-realistic conditions:

Target	Model	RMSE	MAPE

session_count	XGBoost (calendar-only)	4.98	0.350
session_count	Prophet	5.31	0.413
session_count	SARIMA	5.46	0.342
total_energy_kwh	XGBoost (calendar-only)	233.9	0.370
total_energy_kwh	Prophet	262.1	0.449
total_energy_kwh	SARIMA	264.7	0.370

XGBoost (calendar-only) won on RMSE for both targets and on MAPE for energy. SARIMA was marginally ahead on session_count MAPE, a gap too small to change the overall conclusion. We selected XGBoost (calendar-only) as the primary forecasting model. Prophet was kept alongside it for the final forecast because it provides uncertainly intervals that XGBoost does not and because the two models' disagreement about future growth is useful information.

4. The Forecast and Where the Two Models Disagree

Both models were refit on the full historical dataset and used to generate a 30/90/365-day forecast beyond December 2024.

Checking the forecast against the most recent full-year data (2024 averages: 18.0 sessions/hour, 882 kWh/hour) revealed a genuine, structural disagreement between the two models: XGBoost's forecast holds essentially flat at the 2024 level, because a tree-based model has no way to extrapolate beyond the range of years it was trained on.

Prophet's forecast runs roughly 6% higher because it extends the historical growth trend forward.

Neither assumption is verified or provably correct, as they represent two different, reasonable bets on whether the network's recent rapid growth continued into 2025. Rather than resolve this by picking one number, both are shown on the dashboard, so the forecast is used as a planning range, rather than a false point certainty.

5. Dashboard

The Power BI dashboard (two pages for Session count and Energy) presents a historical-vs-forecast view with both models overlaid and a horizon slicer (30/90/365-days), plus weekly and hourly demand-pattern charts, letting a non-technical viewer explore the same findings interactively.

6. Recommendations ● Staff for the midday peak

Demand is concentrated 10 am-2 pm on weekdays, with minimal need for overnight coverage.

● Judge station strain using per-station demand

Since most historical growth reflects new stations rather than existing ones getting busier.

● Use XGBoost's forecast as the working baseline and Prophet's as an upside scenario

Budgeting towards the higher estimate provides a buffer if growth continues, while the flatter estimate represents the more conservative case.

● Resolve the installation-date data issue at the source

Before it affects any project that actual depends on charger lifecycle information.